

HATHEEM RAFEEL

[in](#) Hatheem Rafeel | [✉ hatheemrafeel9999@gmail.com](mailto:hatheemrafeel9999@gmail.com) | [📞 +94 740 487 041](tel:+94740487041) | [🐙 github.com/hatheem-r](https://github.com/hatheem-r) | [🌐 Portfolio](#) | [🌐 Hugging Face](#)

Education

University of Moratuwa, Sri Lanka – BSc in Computer Science and Engineering 2024 – 2028

With Specialization in Data Science and Engineering

- CGPA: 3.53/4.0 (Dean's List in Semester 1: 3.94)

Ak/ Muslim Central College – G.C.E. Advanced Level Examination 2020 – 2023

Physical Science Stream

- Z-Score: 2.3934 | Combined Maths (A), Chemistry (A), Physics (A)

Skills

- **Languages:** Python, JavaScript / TypeScript, SQL
- **ML / Deep Learning:** PyTorch, scikit-learn, XGBoost, LightGBM, CNNs (ResNet-50), BiLSTM-CRF
- **Data Science & Analysis:** pandas, NumPy, Matplotlib, Seaborn, Jupyter, EDA & feature engineering
- **NLP:** Hugging Face (Transformers, Datasets), fastText, SentencePiece
- **LLMs & Agents:** LangGraph, LangChain, LangSmith, RAG, QLoRA fine-tuning, MCP (FastMCP), OpenAI API, Qwen2.5
- **Speech / Audio:** pyannote.audio
- **Data & Backend:** FastAPI, PostgreSQL, Supabase, SQLite, Chroma (vector DB), Pydantic
- **Web & Deployment:** React, Next.js, Node.js / Express, Streamlit, Vercel, Render, Twilio
- **Tools:** Git / GitHub

Projects

SinTOX – Token-Level Sinhala Offensive Language Detection without Pretrained Language Models

CS3631 group project · Ongoing / [🐙 github.com/hatheem-r/project_DNN](https://github.com/hatheem-r/project_DNN)

Tech: Python, PyTorch, Hugging Face Datasets, fastText, SentencePiece, scikit-learn, pandas

- Built a from-scratch BiLSTM-CRF token classifier on the SOLD dataset (10,000 hand-annotated Sinhala tweets) that flags the exact tokens making a tweet offensive, first reproducing the published non-transformer baseline (0.5965 vs 0.60 Macro F1).
- Added a SentencePiece subword channel — words split into pieces, encoded by a per-word BiLSTM and pooled to one vector per token, then concatenated with frozen 300d fastText embeddings — to address the 48.7% of test word types unseen in training caused by Sinhala's agglutinative morphology; lifted offensive-token F1 from 0.597 to 0.708, a precision-neutral gain driven almost entirely by recall (0.498 → 0.677) — all with just 176K trainable parameters and no pretrained embeddings, exceeding published XLM-T (0.70) and landing within 0.012 of XLM-R (0.72) at ~3,180× fewer trainable parameters.

VoxLIT – Learning Interpretability Tool for Voice Models

Semester 5 project · Ongoing / [🐙 github.com/chanuGX/VoxLIT](https://github.com/chanuGX/VoxLIT)

Tech: Python, PyTorch, Hugging Face, pyannote.audio, FastAPI, React

- Extending a LIT-inspired interpretability platform for audio deep learning models by adding three new tasks; owning the speaker diarization module — a pipeline using pretrained pyannote.audio segmentation and speaker-embedding models with clustering to label “who spoke when” on the AMI Meeting Corpus, plus perturbation-based robustness testing and embedding scatter plot / segment similarity heatmap visualizations to interpret model decisions.

MARGENT – Multi-Agent Market-Entry Analysis System for Sri Lanka + Fine-Tuned LLM

Individual project / [🐙 github.com/hatheem-r/project-MARGENT](https://github.com/hatheem-r/project-MARGENT) / [🌐 hatheemr/margent-planner-qwen2.5-3b-lora](https://huggingface.co/hatheemr/margent-planner-qwen2.5-3b-lora)

Tech: Python, LangGraph, LangChain, LangSmith, OpenAI (GPT-4o / GPT-4o-mini), Qwen2.5-3B, QLoRA, Hugging Face, Tavily Search, Pydantic, SQLite, Streamlit

- Architected a multi-agent LangGraph system where parallel research branches (market, pricing, competitors) fan out into a report writer — combining ReAct agents that plan their own Tavily searches and custom tool calls (live USD to LKR API) with a map-reduce competitor subgraph that spawns one parallel worker per discovered competitor via Send() and merges profiles through a state reducer.
- Fine-tuned Qwen2.5-3B with QLoRA to replace GPT-4o-mini at the planner node, distilling ~840 filtered synthetic examples from a teacher model across a forced sector/region/scale/phrasing grid; raised locality-anchored, search-ready research questions from 35% to 100% on a held-out set while schema validity held at 100%, and published the adapter to Hugging Face.
- Implemented a reflection (evaluator-optimizer) loop where a critic agent scores drafts against a grounding rubric using structured Pydantic outputs, two resumable human-in-the-loop interrupt() gates backed by a SQLite check-

pointer, long-term memory recalling a user's past analyses across sessions, and full LangSmith tracing — demoed via a Streamlit UI.

Octwave 2.0 – Machine Learning Competition (IEEE IAS Student Branch Chapter)

Team of two members · Two-round Kaggle competition · Overall Runner-up / github.com/hatheem-r/comp_OCTWAVE

Tech: Python, PyTorch, LightGBM, ResNet-50, scikit-learn, pandas

- Finished as overall competition runner-up across two machine learning challenge rounds; in Round 1 (obesity risk prediction) developed a LightGBM classification model.
- Round 2 — spot-the-difference change detection: designed a two-stream Siamese CNN with a shared pretrained ResNet-50 backbone and feature concatenation to detect added, removed, and changed objects between image pairs; addressed class imbalance and label noise via semantic class merging, and controlled overfitting with high dropout and weight decay — ranked 1st on both the public and private leaderboards for this round.

RAGMAIL – Agentic RAG Assistant over Email

Individual project / github.com/hatheem-r/project-RAGMAIL

Tech: Python, LangChain, LangGraph, MCP (FastMCP), Chroma, OpenAI (GPT-4o-mini, text-embedding-3-small), IMAP, Streamlit

- Built a multimodal RAG pipeline — chunking, OpenAI embeddings, and a persistent Chroma vector store — that transcribes image attachments through a vision model at ingest time so their contents become semantically searchable, with retrieved chunks carrying sender/date/subject metadata to force source-cited, grounded answers.
- Developed an MCP (Model Context Protocol) server with FastMCP over streamable HTTP exposing tools, resources, and prompts for live inbox access, and wired it into a LangGraph tool-calling agent that routes deliberately between MMR-based retrieval and live MCP lookups, backed by per-thread short-term memory (checkpointer) and vector-based long-term memory.

aPOINT – Multi-Agent AI System for Hotel Operations

Individual project / github.com/hatheem-r/project-APOINT

Tech: Python, FastAPI, LangGraph, LangChain, OpenAI (GPT-4o-mini), Supabase (Postgres + Auth), Next.js, Twilio, Vercel, Render

- Built a LangGraph multi-agent backend where a routing “boss” agent classifies each guest message and dispatches to specialist agents — booking, worker-dispatch, and concierge — that execute OpenAI tool-calling loops against live Supabase Postgres tools (room availability, booking, staff-on-shift lookup, task assignment), with conversation memory replaying recent exchanges so multi-turn requests resolve correctly.
- Unified three ingestion channels (web, WhatsApp/Twilio, email) into one normalized agent pipeline with identity-based room resolution, and enforced deterministic guardrails in code rather than prompts — per-request counters preventing duplicate bookings/assignments, service gating on active bookings, and blocked disclosure of internal data; deployed on Render/Vercel with a Next.js role-gated frontend and Supabase Auth JWT verification.

EazyRents – Full-stack vehicle rental platform for tourists in Sri Lanka

Individual project / github.com/hatheem-r/EAZYRENTS

Tech: React, Node.js/Express, PostgreSQL, OpenAI API

- Built a peer-to-peer booking system with JWT auth, role-based access, and double-booking prevention enforced at the database layer via PostgreSQL exclusion constraints while preserving ACID properties throughout the database; integrated a domain-scoped, prompt-engineered LLM travel assistant (GPT-4o-mini) with stateless conversation handling, schema-validated inputs, and per-IP rate limiting for vehicle and itinerary recommendations.

More projects

- **Customer Churn Prediction** — classical ML pipeline in scikit-learn (EDA, preprocessing, class-imbalance handling) comparing Logistic Regression, Random Forest, and XGBoost on precision / recall / F1 / ROC-AUC.
github.com/hatheem-r/Customer_Churn_Pred

Certificates

- Agentic AI — DeepLearning.AI by Andrew Ng
- Supervised Machine Learning: Regression and Classification — DeepLearning.AI and Stanford University by Andrew Ng
- Foundation: Introduction to LangGraph — Python by LangChain Academy
- LangChain Chat with your Data — LangChain by Harrison Chase
- LangChain for LLM Application Development — LangChain by Harrison Chase, Andrew Ng
- Intro to Model Context Protocol (MCP) at Scrimba

Achievements & Awards

- Runner-up, Octwave 2.0 Machine Learning Competition (IEEE IAS Student Branch Chapter), 2025 — placed 1st on the public and private leaderboards in the change-detection round.

Experiences & Extracurriculars

- IEEE CIS SB UOM Chapter — Treasurer
- Events Committee — SLIoT, 2025
- Programme Committee — RoboGames, 2025
- MoraSpirit — Web & Technology Pillar

References

1. Prof. Dulani Meedeniya — Senior Lecturer, Department of Computer Science & Engineering, Faculty of Engineering, University of Moratuwa, Sri Lanka. Email: dulanim@cse.mrt.ac.lk